# MSVTNet: Multi-Scale Vision Transformer Neural Network for EEG-Based Motor Imagery Decoding

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## I. Single Subject Classification Accuracies

For reproducibility and future comparisons, the single subject MI classification accuracies achieved by FBCSP-SVM [1], Shallow ConvNet [2], EEGNet [3], FBCNet [4], Ligth ConvNet [5] and MSVTNet for subject-dependent (session-dependent and session-independent) and subject-independent analyses on BCI Competition IV 2a (BCIC-IV-2a) [6], BCI Competition IV 2b (BCIC-IV-2b) [7] and OpenBMI [8] datasets are presented in Table I, II, III, IV and V, respectively.

TABLE I
Classification Accuracies in % for Each Subject in BCIC-IV-2a Dataset

| Experiment | | Method | S01 | S02 | S03 | S04 | S05 | S06 | S07 | S08 | S09 | Avg | Std |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Subject-dependent | Session-dependent | FBCSP-SVM | 79.17 | 65.28 | 88.89 | 62.50 | 70.83 | 43.06 | 93.06 | 84.72 | 63.89 | 72.38 | 14.84 |
| | | Shallow ConvNet | 74.17 | 63.33 | 90.56 | 60.56 | 65.56 | 66.94 | 80.56 | 80.00 | 78.33 | 73.33 | 9.33 |
| | | EEGNet | 75.28 | 52.50 | 88.33 | 69.72 | 69.72 | 67.50 | 79.44 | 85.00 | 88.61 | 75.12 | 11.06 |
| | | FBCNet | 78.06 | 56.94 | 95.28 | 68.33 | 79.17 | 50.28 | 84.44 | 87.50 | 78.89 | 75.43 | 13.66 |
| | | Light ConvNet | 84.17 | 62.78 | 91.11 | 73.61 | 80.28 | 61.94 | 92.50 | 89.72 | 79.17 | 79.48 | 10.82 |
| | | MSVTNet | 81.67 | 64.17 | 93.61 | 77.78 | 80.56 | 78.61 | 90.56 | 87.50 | 88.61 | 82.56 | 8.36 |
| | Session-independent | FBCSP-SVM | 81.60 | 50.00 | 80.90 | 60.07 | 46.53 | 46.53 | 85.07 | 78.13 | 70.14 | 66.56 | 15.06 |
| | | Shallow ConvNet | 74.38 | 56.46 | 84.38 | 68.75 | 54.24 | 55.83 | 74.38 | 81.32 | 71.88 | 69.07 | 10.56 |
| | | EEGNet | 81.74 | 52.92 | 85.42 | 63.19 | 70.62 | 56.81 | 84.24 | 78.12 | 80.62 | 72.63 | 11.58 |
| | | FBCNet | 81.88 | 54.44 | 89.31 | 76.94 | 65.42 | 55.90 | 84.72 | 79.24 | 81.94 | 74.42 | 11.99 |
| | | Light ConvNet | 86.46 | 54.31 | 90.49 | 78.47 | 64.10 | 60.35 | 86.25 | 84.65 | 84.31 | 76.60 | 12.60 |
| | | MSVTNet | 85.76 | 54.38 | 91.74 | 75.90 | 77.22 | 68.26 | 88.26 | 80.14 | 84.79 | 78.50 | 10.86 |
| Subject-independent | | FBCSP-SVM | 41.15 | 25.00 | 53.30 | 31.25 | 27.43 | 28.65 | 22.74 | 56.25 | 43.23 | 36.56 | 11.70 |
| | | Shallow ConvNet | 66.67 | 39.93 | 81.08 | 50.87 | 44.62 | 49.65 | 68.58 | 73.96 | 67.19 | 60.28 | 13.49 |
| | | EEGNet | 72.57 | 47.40 | 75.87 | 56.08 | 55.03 | 44.79 | 65.97 | 75.52 | 66.15 | 62.15 | 11.12 |
| | | FBCNet | 67.01 | 37.50 | 65.97 | 50.87 | 27.43 | 32.47 | 43.75 | 63.54 | 58.68 | 49.69 | 14.20 |
| | | Light ConvNet | 64.58 | 34.72 | 68.23 | 46.53 | 37.33 | 35.76 | 44.97 | 61.28 | 57.99 | 50.15 | 12.34 |
| | | MSVTNet | 67.71 | 45.31 | 85.76 | 53.30 | 50.35 | 47.40 | 75.17 | 81.42 | 70.49 | 64.10 | 14.47 |

TABLE II
Classification Accuracies in % for Each Subject in BCIC-IV-2b Dataset

| Experiment | | Method | S01 | S02 | S03 | S04 | S05 | S06 | S07 | S08 | S09 | Avg | Std |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Subject-dependent | Session-dependent | FBCSP-SVM | 56.67 | 50.00 | 66.67 | 80.00 | 88.57 | 70.00 | 60.00 | 63.33 | 36.67 | 63.54 | 14.58 |
| | | Shallow ConvNet | 56.00 | 57.33 | 43.33 | 86.29 | 83.43 | 78.67 | 65.33 | 74.00 | 44.00 | 65.38 | 15.31 |
| | | EEGNet | 65.33 | 62.67 | 50.67 | 87.43 | 69.71 | 91.33 | 78.67 | 62.67 | 54.00 | 69.16 | 13.28 |
| | | FBCNet | 58.00 | 55.33 | 55.33 | 93.14 | 90.86 | 85.33 | 52.67 | 64.00 | 46.00 | 66.74 | 16.99 |
| | | Light ConvNet | 64.00 | 62.00 | 51.33 | 92.57 | 88.00 | 76.67 | 66.00 | 65.33 | 46.67 | 68.06 | 14.43 |
| | | MSVTNet | 60.67 | 67.33 | 58.00 | 90.86 | 81.14 | 82.00 | 78.67 | 62.67 | 51.33 | 70.30 | 12.55 |
| | Session-independent | FBCSP-SVM | 62.19 | 56.43 | 52.19 | 95.31 | 86.25 | 79.69 | 77.19 | 87.19 | 77.81 | 74.92 | 13.95 |
| | | Shallow ConvNet | 71.38 | 65.21 | 79.00 | 81.81 | 81.31 | 79.81 | 83.37 | 90.56 | 82.69 | 79.46 | 6.88 |
| | | EEGNet | 74.56 | 67.36 | 84.50 | 97.25 | 91.31 | 78.94 | 93.06 | 93.56 | 88.38 | 85.44 | 9.40 |
| | | FBCNet | 71.94 | 51.36 | 64.19 | 96.50 | 91.50 | 82.94 | 81.44 | 91.75 | 86.56 | 79.80 | 13.87 |
| | | Light ConvNet | 74.38 | 54.21 | 69.12 | 97.31 | 91.81 | 86.19 | 83.00 | 91.44 | 84.50 | 81.33 | 12.63 |
| | | MSVTNet | 75.56 | 70.36 | 85.00 | 96.06 | 94.69 | 85.06 | 91.50 | 94.06 | 88.00 | 86.70 | 8.35 |
| Subject-independent | | FBCSP-SVM | 70.97 | 54.56 | 52.50 | 74.86 | 67.30 | 75.56 | 73.75 | 66.18 | 73.61 | 67.80 | 8.17 |
| | | Shallow ConvNet | 73.47 | 66.76 | 60.56 | 80.81 | 82.16 | 79.44 | 78.19 | 75.00 | 72.22 | 74.29 | 6.64 |
| | | EEGNet | 76.53 | 70.44 | 64.72 | 75.68 | 83.65 | 76.81 | 83.75 | 79.47 | 73.61 | 76.07 | 5.73 |
| | | FBCNet | 74.03 | 59.12 | 61.67 | 80.54 | 78.78 | 73.33 | 70.97 | 74.34 | 76.94 | 72.19 | 6.90 |
| | | Light ConvNet | 71.25 | 59.41 | 61.94 | 83.78 | 81.89 | 74.03 | 72.36 | 72.63 | 75.42 | 72.52 | 7.53 |
| | | MSVTNet | 74.86 | 71.47 | 61.94 | 82.03 | 83.24 | 85.00 | 81.67 | 71.45 | 76.39 | 76.45 | 7.00 |

## II. Confusion Matrix for All Experiments

To more comprehensively compare the performance of different decoding algorithms on the BCIC-IV-2a, 2b and OpenBMI datasets, the confusion matrices of all analysis experiments are displayed in Fig. 1, 2 and 3, respectively. Compared with the contrastive algorithms, MSVTNet achieved higher decoding accuracy for each MI class with smaller variance in all analysis experiments, demonstrating the reliability and generalization of the model decoding performance.

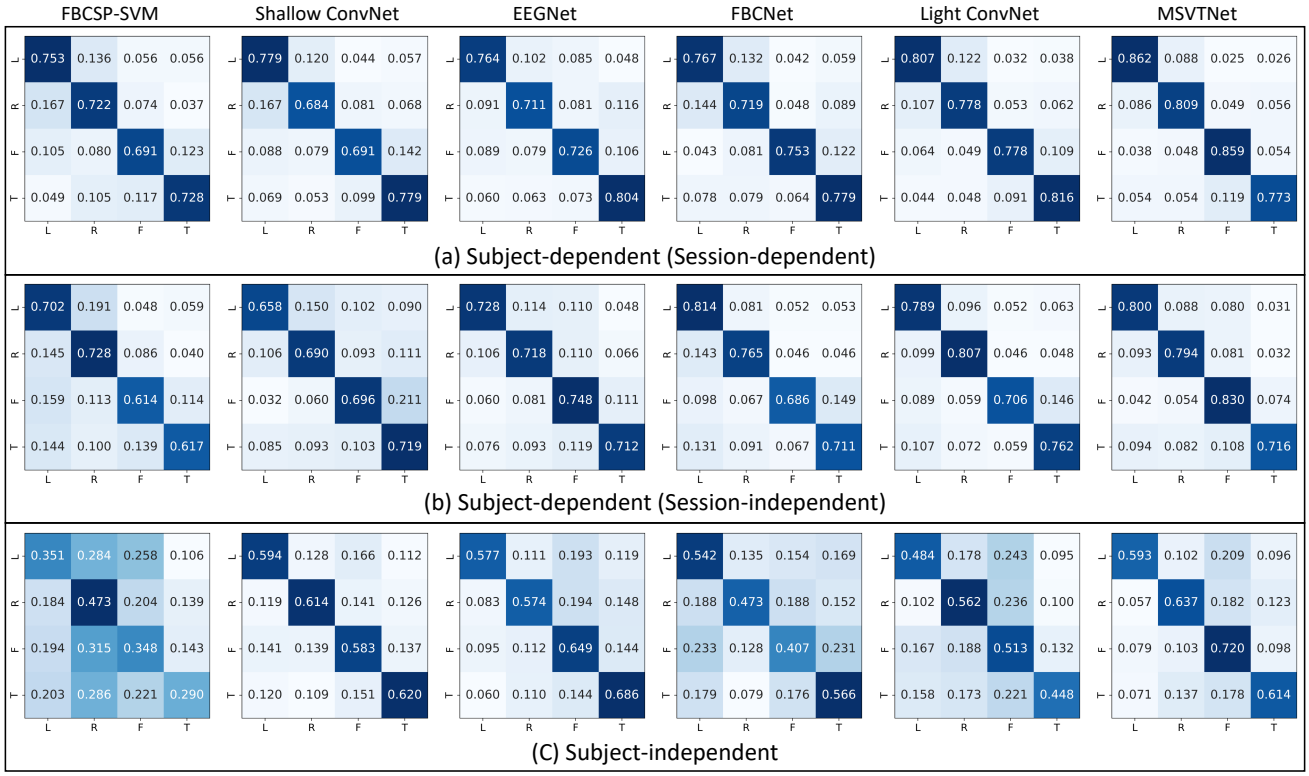


Fig. 1. Confusion matrices for different models on the BCIC-IV-2a dataset. Each column represents the true label and each row depicts the predicted label. L, R, F, and T refer to MI of left hand, right hand, feet and tongue, respectively.


[1] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter bank common spatial pattern (fbcsp) in brain-computer interface," in *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*, pp. 2390–2397, IEEE, 2008.

[2] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

[3] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

[4] R. Mane, E. Chew, K. Chua, K. K. Ang, N. Robinson, A. P. Vinod, S.-W. Lee, and C. Guan, "FBCNet: A multi-view convolutional neural network for brain-computer interface," *arXiv preprint arXiv:2104.01233*, 2021.

[5] X. Ma, W. Chen, Z. Pei, J. Liu, B. Huang, and J. Chen, "A temporal dependency learning CNN with attention mechanism for MI-EEG decoding," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023.

[6] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "BCI Competition 2008–Graz data set A," *Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology*, vol. 16, pp. 1–6, 2008.

[7] R. Leeb, C. Brunner, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "BCI Competition 2008–Graz data set B," *Graz University of Technology, Austria*, vol. 16, pp. 1–6, 2008.

[8] M.-H. Lee, O.-Y. Kwon, Y.-J. Kim, H.-K. Kim, Y.-E. Lee, J. Williamson, S. Fazli, and S.-W. Lee, "EEG dataset and OpenBMI toolbox for three BCI paradigms: An investigation into BCI illiteracy," *GigaScience*, vol. 8, no. 5, p. giz002, 2019.

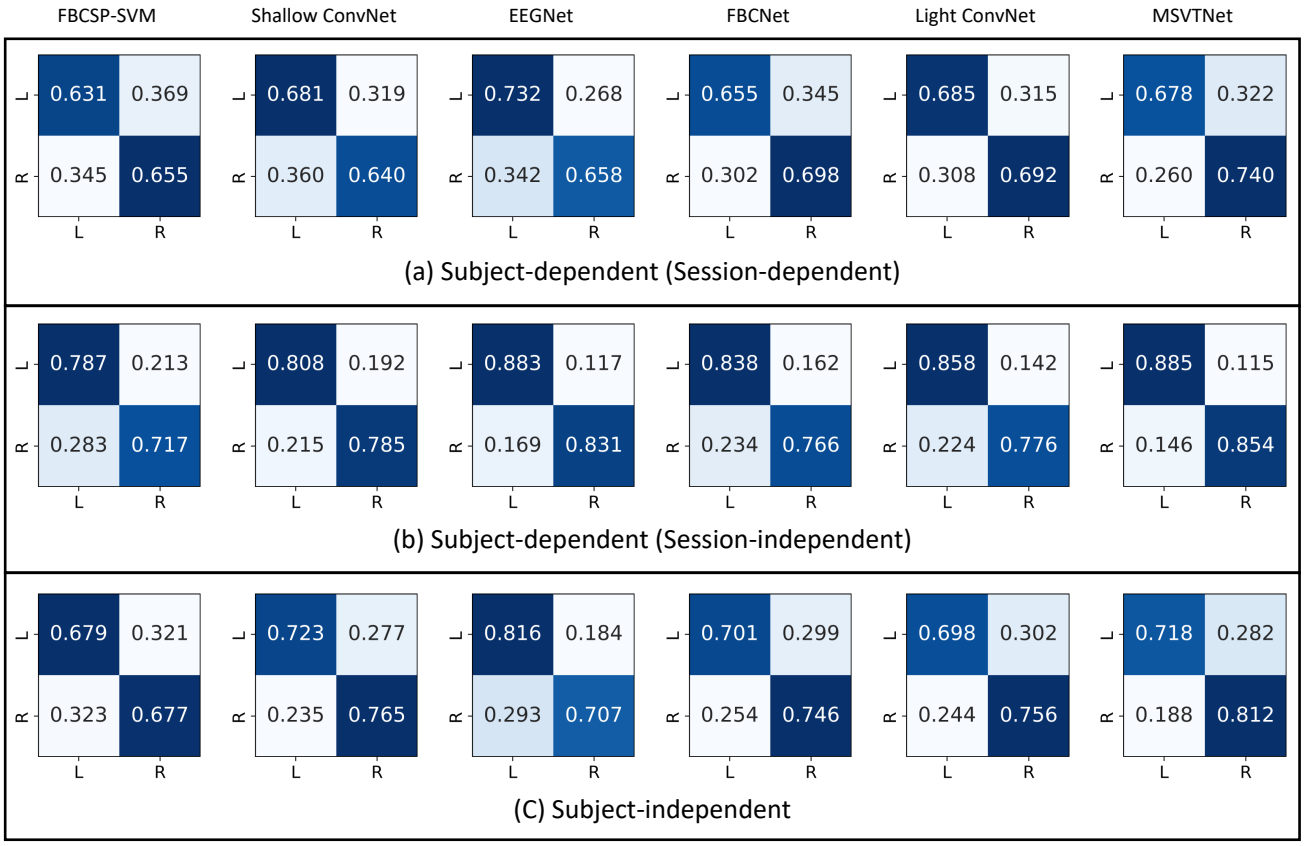


Fig. 2. Confusion matrices for different models on the BCIC-IV-2b dataset. Each column represents the true label and each row depicts the predicted label. L and R refer to MI of left hand and right hand, respectively.

TABLE III
CLASSIFICATION ACCURACIES IN % USING SUBJECT-DEPENDENT (SESSION-DEPENDENT) ANALYSIS FOR EACH SUBJECT IN OPENBMI DATASET

| Subject No. | FBCSP-SVM | Shallow ConvNet | EEGNet | FBCNet | Light ConvNet | MSVTNet |
|---|---|---|---|---|---|---|
| 1 | 78.00 | 74.80 | 80.80 | 81.20 | 74.00 | 82.40 |
| 2 | 100.00 | 98.40 | 100.00 | 100.00 | 98.80 | 100.00 |
| 3 | 90.00 | 93.20 | 97.20 | 94.40 | 87.20 | 97.60 |
| 4 | 58.00 | 62.80 | 65.20 | 66.40 | 60.40 | 63.60 |
| 5 | 98.00 | 85.60 | 98.00 | 91.20 | 96.80 | 98.80 |
| 6 | 78.00 | 70.00 | 79.60 | 76.80 | 80.00 | 81.20 |
| 7 | 60.00 | 53.60 | 49.60 | 56.80 | 57.60 | 66.80 |
| 8 | 40.00 | 55.20 | 69.20 | 78.40 | 69.60 | 67.20 |
| 9 | 78.00 | 76.80 | 85.20 | 79.60 | 83.20 | 80.00 |
| 10 | 64.00 | 69.20 | 71.20 | 76.00 | 70.40 | 74.40 |
| 11 | 42.00 | 50.00 | 46.00 | 52.80 | 60.40 | 58.80 |
| 12 | 56.00 | 63.60 | 73.20 | 66.00 | 73.60 | 75.20 |
| 13 | 54.00 | 44.80 | 52.40 | 56.80 | 61.60 | 70.80 |
| 14 | 68.00 | 58.80 | 57.60 | 74.40 | 70.00 | 70.40 |
| 15 | 50.00 | 63.20 | 72.40 | 76.80 | 71.20 | 76.80 |
| 16 | 42.00 | 61.20 | 58.80 | 58.40 | 53.20 | 54.40 |
| 17 | 78.00 | 75.60 | 79.60 | 82.40 | 83.20 | 88.40 |
| 18 | 90.00 | 84.00 | 94.00 | 86.00 | 91.20 | 90.80 |
| 19 | 76.00 | 84.40 | 83.60 | 88.00 | 80.80 | 83.60 |
| 20 | 52.00 | 52.80 | 56.40 | 60.40 | 70.40 | 57.60 |
| 21 | 98.00 | 96.00 | 98.40 | 98.40 | 99.60 | 100.00 |
| 22 | 90.00 | 76.40 | 90.40 | 90.80 | 89.60 | 89.60 |
| 23 | 68.00 | 64.80 | 69.20 | 67.20 | 65.20 | 73.20 |
| 24 | 52.00 | 56.80 | 47.20 | 47.60 | 66.00 | 63.60 |
| 25 | 52.00 | 60.80 | 55.60 | 68.80 | 64.40 | 62.40 |
| 26 | 52.00 | 58.00 | 49.20 | 55.60 | 52.40 | 53.20 |
| 27 | 52.00 | 57.60 | 50.40 | 66.80 | 63.20 | 57.20 |
| 28 | 98.00 | 89.60 | 92.80 | 94.00 | 94.40 | 94.40 |
| 29 | 94.00 | 84.00 | 86.80 | 90.80 | 92.00 | 86.80 |
| 30 | 82.00 | 63.60 | 69.20 | 74.80 | 74.80 | 75.60 |
| 31 | 76.00 | 70.00 | 76.40 | 74.40 | 76.40 | 85.60 |
| 32 | 78.00 | 84.80 | 79.20 | 88.40 | 83.20 | 85.20 |
| 33 | 100.00 | 94.00 | 99.60 | 100.00 | 98.40 | 100.00 |
| 34 | 42.00 | 55.20 | 59.20 | 67.20 | 59.60 | 60.40 |
| 35 | 64.00 | 83.60 | 86.80 | 85.20 | 88.80 | 89.60 |
| 36 | 96.00 | 92.00 | 96.00 | 92.40 | 94.40 | 96.40 |
| 37 | 98.00 | 90.80 | 95.60 | 96.00 | 94.80 | 98.00 |
| 38 | 46.00 | 60.80 | 49.20 | 61.60 | 56.40 | 57.20 |
| 39 | 92.00 | 84.00 | 88.00 | 84.80 | 89.20 | 95.20 |
| 40 | 58.00 | 50.00 | 60.40 | 54.00 | 54.00 | 60.80 |
| 41 | 66.00 | 59.60 | 55.20 | 66.00 | 65.60 | 71.60 |
| 42 | 46.00 | 50.00 | 50.00 | 57.20 | 58.40 | 54.00 |
| 43 | 86.00 | 71.20 | 77.20 | 77.20 | 82.80 | 82.00 |
| 44 | 98.00 | 88.80 | 97.20 | 94.00 | 97.20 | 98.80 |
| 45 | 92.00 | 87.60 | 88.80 | 90.00 | 92.40 | 92.00 |
| 46 | 56.00 | 58.40 | 48.80 | 58.00 | 70.00 | 60.80 |
| 47 | 48.00 | 52.40 | 46.00 | 51.20 | 59.60 | 50.40 |
| 48 | 56.00 | 70.00 | 64.00 | 75.20 | 83.20 | 74.40 |
| 49 | 70.00 | 63.60 | 73.20 | 75.60 | 74.40 | 76.00 |
| 50 | 50.00 | 51.20 | 57.20 | 55.60 | 62.80 | 59.20 |
| 51 | 58.00 | 52.40 | 64.00 | 69.20 | 66.80 | 68.80 |
| 52 | 78.00 | 63.20 | 72.80 | 72.40 | 73.20 | 76.80 |
| 53 | 52.00 | 51.60 | 55.20 | 63.20 | 62.80 | 58.00 |
| 54 | 66.00 | 50.80 | 50.40 | 55.60 | 51.60 | 54.00 |
| Avg | 69.67 | 68.92 | 71.66 | 74.48 | 75.02 | 75.93 |
| Std | 18.82 | 14.90 | 17.31 | 14.52 | 14.16 | 15.14 |

TABLE IV
CLASSIFICATION ACCURACIES IN % USING SUBJECT-DEPENDENT (SESSION-INDEPENDENT) ANALYSIS FOR EACH SUBJECT IN OPENBMI DATASET

| Subject No. | FBCSP-SVM | Shallow ConvNet | EEGNet | FBCNet | Light ConvNet | MSVTNet |
|---|---|---|---|---|---|---|
| 1 | 81.00 | 80.50 | 88.50 | 87.70 | 87.40 | 86.40 |
| 2 | 63.00 | 72.50 | 72.40 | 64.90 | 61.00 | 68.30 |
| 3 | 96.00 | 94.40 | 96.30 | 96.20 | 95.50 | 96.80 |
| 4 | 68.00 | 61.50 | 63.90 | 76.70 | 78.40 | 74.50 |
| 5 | 89.50 | 83.80 | 84.60 | 87.60 | 89.60 | 91.00 |
| 6 | 84.50 | 75.00 | 84.30 | 82.20 | 83.10 | 84.80 |
| 7 | 71.50 | 54.30 | 57.70 | 65.90 | 73.00 | 75.10 |
| 8 | 57.50 | 61.40 | 65.80 | 65.20 | 62.60 | 63.20 |
| 9 | 71.00 | 72.80 | 81.60 | 79.90 | 81.90 | 81.90 |
| 10 | 53.00 | 58.90 | 60.30 | 62.70 | 66.20 | 61.10 |
| 11 | 50.00 | 53.60 | 51.60 | 52.10 | 54.50 | 58.30 |
| 12 | 61.00 | 65.60 | 62.30 | 69.90 | 67.50 | 71.20 |
| 13 | 59.50 | 56.70 | 56.80 | 54.70 | 57.30 | 58.50 |
| 14 | 54.00 | 56.40 | 54.40 | 59.00 | 59.80 | 62.90 |
| 15 | 47.50 | 60.60 | 59.40 | 59.60 | 57.40 | 55.00 |
| 16 | 55.50 | 50.10 | 50.30 | 57.40 | 60.10 | 57.90 |
| 17 | 49.50 | 66.10 | 52.70 | 58.00 | 52.60 | 62.10 |
| 18 | 72.50 | 82.50 | 88.70 | 87.00 | 91.00 | 87.90 |
| 19 | 73.00 | 75.70 | 75.80 | 69.30 | 77.10 | 71.20 |
| 20 | 64.50 | 60.40 | 59.40 | 71.90 | 74.20 | 78.40 |
| 21 | 99.50 | 96.60 | 99.70 | 99.20 | 99.60 | 98.60 |
| 22 | 52.00 | 66.50 | 74.90 | 67.70 | 73.40 | 77.20 |
| 23 | 66.00 | 66.30 | 60.30 | 66.10 | 68.90 | 67.90 |
| 24 | 49.00 | 50.70 | 51.10 | 52.00 | 53.90 | 57.30 |
| 25 | 59.00 | 68.30 | 62.50 | 66.70 | 65.70 | 64.50 |
| 26 | 61.00 | 53.30 | 49.60 | 54.50 | 53.60 | 55.20 |
| 27 | 57.50 | 54.30 | 54.10 | 55.90 | 58.90 | 54.40 |
| 28 | 95.00 | 88.10 | 91.50 | 95.70 | 97.90 | 95.00 |
| 29 | 97.00 | 93.30 | 94.60 | 93.50 | 95.00 | 90.40 |
| 30 | 65.00 | 61.90 | 65.00 | 68.80 | 71.20 | 72.90 |
| 31 | 71.00 | 59.20 | 59.80 | 67.20 | 71.00 | 68.70 |
| 32 | 92.50 | 87.00 | 87.50 | 90.10 | 85.70 | 92.70 |
| 33 | 93.00 | 95.30 | 96.00 | 97.20 | 97.40 | 96.30 |
| 34 | 52.50 | 52.40 | 51.20 | 55.30 | 58.00 | 54.60 |
| 35 | 58.00 | 70.80 | 68.80 | 73.60 | 70.20 | 79.60 |
| 36 | 98.50 | 96.60 | 97.50 | 97.00 | 98.40 | 97.70 |
| 37 | 93.50 | 92.50 | 96.60 | 95.20 | 94.10 | 95.40 |
| 38 | 61.00 | 49.50 | 53.00 | 57.40 | 61.90 | 58.30 |
| 39 | 77.00 | 79.20 | 84.00 | 83.00 | 84.20 | 85.20 |
| 40 | 58.50 | 56.90 | 57.00 | 58.00 | 58.20 | 63.60 |
| 41 | 60.50 | 49.40 | 54.20 | 57.90 | 59.50 | 58.90 |
| 42 | 64.00 | 54.70 | 55.30 | 65.30 | 66.00 | 64.40 |
| 43 | 62.00 | 69.00 | 77.40 | 76.40 | 81.60 | 85.90 |
| 44 | 98.00 | 96.10 | 98.80 | 97.90 | 97.50 | 97.20 |
| 45 | 94.50 | 90.10 | 93.70 | 91.80 | 93.80 | 95.60 |
| 46 | 62.00 | 61.20 | 61.10 | 74.60 | 70.00 | 64.70 |
| 47 | 49.00 | 56.70 | 53.70 | 60.50 | 64.20 | 56.80 |
| 48 | 51.50 | 57.90 | 60.00 | 62.40 | 63.30 | 64.90 |
| 49 | 58.50 | 67.20 | 69.10 | 63.60 | 70.50 | 71.50 |
| 50 | 49.00 | 51.70 | 52.30 | 55.60 | 55.70 | 53.40 |
| 51 | 58.50 | 58.50 | 53.40 | 58.80 | 59.80 | 54.70 |
| 52 | 80.00 | 67.10 | 79.90 | 76.70 | 78.50 | 76.20 |
| 53 | 60.00 | 53.70 | 55.50 | 54.90 | 58.20 | 56.10 |
| 54 | 57.00 | 53.20 | 54.80 | 58.60 | 57.60 | 56.40 |
| Avg | 68.20 | 67.93 | 69.27 | 71.43 | 72.66 | 72.75 |
| Std | 15.95 | 14.68 | 16.21 | 14.58 | 14.48 | 14.66 |

TABLE V
CLASSIFICATION ACCURACIES IN % USING SUBJECT-INDEPENDENT ANALYSIS FOR EACH SUBJECT IN OPENBMI DATASET

| Subject No. | FBCSP-SVM | Shallow ConvNet | EEGNet | FBCNet | Light ConvNet | MSVTNet |
|---|---|---|---|---|---|---|
| 1 | 79.50 | 66.17 | 84.00 | 82.25 | 79.75 | 85.25 |
| 2 | 55.00 | 86.50 | 81.00 | 74.75 | 80.25 | 86.50 |
| 3 | 87.75 | 97.67 | 95.50 | 93.00 | 94.50 | 97.75 |
| 4 | 61.25 | 73.67 | 55.75 | 51.50 | 71.25 | 74.75 |
| 5 | 74.50 | 90.17 | 95.75 | 89.00 | 92.25 | 92.50 |
| 6 | 80.25 | 83.00 | 89.00 | 81.00 | 84.00 | 88.75 |
| 7 | 52.75 | 60.17 | 66.50 | 63.00 | 65.75 | 73.50 |
| 8 | 67.25 | 68.17 | 76.00 | 71.75 | 69.00 | 76.50 |
| 9 | 80.25 | 82.50 | 85.25 | 81.75 | 83.50 | 82.50 |
| 10 | 53.25 | 66.00 | 60.75 | 59.75 | 62.25 | 60.50 |
| 11 | 50.25 | 60.00 | 62.00 | 56.00 | 59.25 | 65.75 |
| 12 | 58.50 | 69.00 | 73.75 | 72.00 | 72.25 | 78.00 |
| 13 | 56.75 | 57.83 | 67.25 | 61.50 | 54.00 | 65.25 |
| 14 | 58.50 | 75.00 | 81.25 | 73.75 | 75.25 | 80.25 |
| 15 | 54.50 | 54.00 | 62.25 | 56.25 | 56.75 | 62.50 |
| 16 | 65.00 | 70.50 | 71.25 | 70.75 | 70.25 | 68.00 |
| 17 | 64.25 | 68.50 | 78.25 | 75.00 | 74.25 | 83.25 |
| 18 | 75.50 | 84.67 | 91.75 | 83.75 | 83.50 | 89.75 |
| 19 | 69.50 | 83.67 | 83.25 | 77.75 | 77.50 | 82.00 |
| 20 | 72.00 | 74.17 | 80.25 | 75.50 | 76.75 | 81.25 |
| 21 | 83.75 | 79.00 | 92.00 | 87.00 | 86.50 | 92.50 |
| 22 | 78.00 | 74.67 | 84.50 | 77.50 | 84.00 | 85.50 |
| 23 | 67.00 | 79.67 | 70.50 | 71.00 | 69.75 | 73.50 |
| 24 | 49.75 | 55.00 | 54.00 | 55.25 | 54.75 | 58.00 |
| 25 | 59.75 | 68.67 | 68.75 | 73.25 | 63.75 | 79.00 |
| 26 | 56.25 | 61.33 | 67.25 | 64.75 | 64.00 | 68.00 |
| 27 | 59.25 | 58.67 | 66.25 | 61.75 | 62.00 | 70.75 |
| 28 | 91.50 | 94.17 | 98.75 | 96.00 | 97.25 | 97.00 |
| 29 | 85.25 | 49.50 | 66.50 | 57.50 | 57.75 | 56.25 |
| 30 | 72.50 | 71.00 | 75.75 | 74.75 | 70.75 | 80.75 |
| 31 | 73.00 | 60.83 | 75.25 | 76.00 | 76.25 | 78.50 |
| 32 | 84.50 | 86.50 | 85.25 | 81.75 | 86.00 | 87.75 |
| 33 | 93.75 | 98.33 | 98.00 | 96.00 | 97.75 | 97.00 |
| 34 | 50.00 | 49.50 | 54.25 | 54.00 | 53.50 | 57.50 |
| 35 | 48.00 | 69.67 | 58.50 | 60.75 | 65.00 | 68.75 |
| 36 | 84.50 | 99.00 | 95.25 | 94.50 | 90.00 | 96.50 |
| 37 | 94.75 | 96.83 | 97.00 | 95.25 | 96.50 | 95.50 |
| 38 | 59.25 | 61.50 | 64.50 | 56.50 | 64.25 | 66.50 |
| 39 | 85.00 | 86.83 | 90.25 | 87.50 | 86.00 | 88.25 |
| 40 | 60.00 | 70.83 | 68.25 | 64.00 | 64.75 | 74.50 |
| 41 | 51.50 | 59.00 | 56.25 | 55.00 | 66.50 | 67.75 |
| 42 | 65.50 | 73.33 | 77.50 | 72.00 | 69.75 | 75.75 |
| 43 | 68.50 | 88.17 | 86.25 | 86.25 | 85.50 | 85.25 |
| 44 | 89.25 | 93.33 | 94.25 | 91.75 | 92.75 | 95.25 |
| 45 | 80.00 | 84.50 | 88.75 | 86.50 | 91.00 | 85.75 |
| 46 | 69.50 | 64.50 | 77.75 | 75.50 | 70.00 | 77.25 |
| 47 | 65.00 | 64.67 | 73.75 | 73.00 | 72.00 | 73.50 |
| 48 | 52.25 | 62.17 | 53.50 | 61.50 | 59.50 | 64.25 |
| 49 | 61.50 | 55.17 | 67.50 | 69.50 | 69.75 | 70.50 |
| 50 | 52.00 | 53.67 | 53.50 | 55.00 | 52.25 | 58.75 |
| 51 | 60.25 | 64.17 | 67.50 | 64.75 | 68.00 | 67.50 |
| 52 | 69.00 | 81.17 | 82.75 | 77.00 | 76.00 | 80.25 |
| 53 | 57.75 | 60.50 | 65.25 | 58.00 | 60.25 | 70.75 |
| 54 | 53.25 | 62.33 | 62.75 | 60.50 | 64.00 | 61.25 |
| Avg | 67.55 | 72.40 | 75.53 | 72.62 | 73.52 | 77.41 |
| Std | 13.03 | 13.48 | 13.19 | 12.62 | 12.39 | 11.53 |

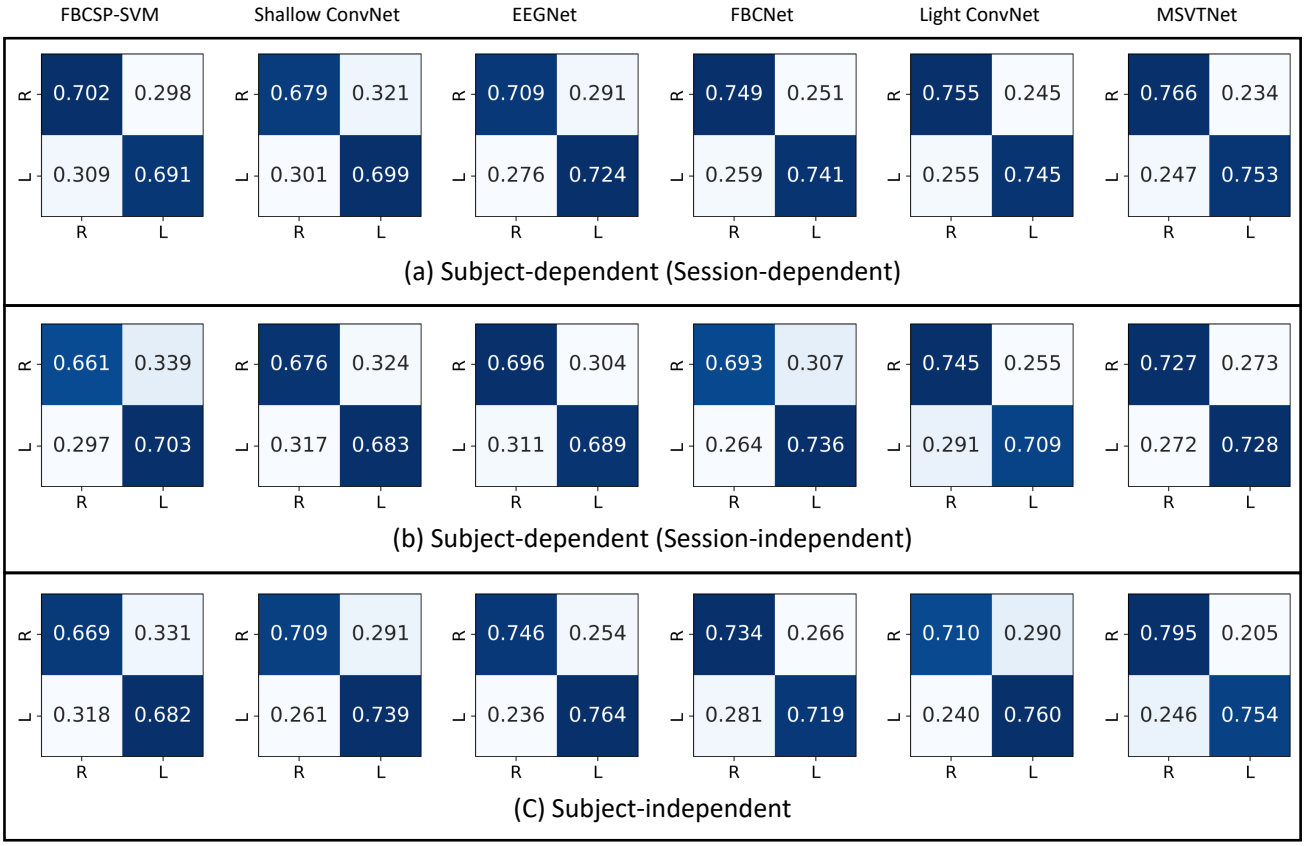


Fig. 3. Confusion matrices for different models on the OpenBMI dataset. Each column represents the true label and each row depicts the predicted label. L and R refer to MI of left hand and right hand, respectively.